

Classical Mechanics

Complete Study Guide for GATE 2026

Problem-Solving Intensive Mastery

Comprehensive Coverage of All Topics

Prepared: December 2025

Contents

1	Introduction	3
1.1	Why Classical Mechanics is Crucial	3
1.2	Study Strategy	3
2	Lagrangian Formulation	4
2.1	Motivation and Fundamental Concepts	4
2.2	Generalized Coordinates	4
2.3	D'Alembert's Principle and Virtual Work	4
2.4	Euler-Lagrange Equations	5
2.5	Constraints	6
2.6	Lagrange Multipliers Method	6
2.7	Generalized Momentum and Cyclic Coordinates	7
2.8	Conservation Laws and Noether's Theorem	8
2.9	Energy Function and Conservation	8
2.10	Practice Problems - Lagrangian Mechanics	8
3	Hamiltonian Mechanics	9
3.1	From Lagrangian to Hamiltonian	9
3.2	Hamilton's Equations	9
3.3	Phase Space	10
3.4	Poisson Brackets	10
3.5	Properties of Poisson Brackets	11
3.6	Canonical Transformations	11
3.7	Generating Functions	12
3.8	Hamilton-Jacobi Theory	12
3.9	Action-Angle Variables	13
3.10	Practice Problems - Hamiltonian Mechanics	13
4	Central Force Motion	14
4.1	Central Force Definition	14
4.2	Conservation Laws	14
4.3	Equation of Motion in Plane	14
4.4	Effective Potential	15
4.5	Orbit Equation	15
4.6	Kepler Problem (Inverse Square Law)	15
4.6.1	Orbit Equation for Kepler Problem	15
4.6.2	Kepler's Laws	16
4.7	Energy Analysis	16
4.8	Scattering in Central Force	17
4.9	Virial Theorem	17
4.10	Practice Problems - Central Force Motion	17
5	Rigid Body Dynamics	18
5.1	Rigid Body Definition and Kinematics	18
5.2	Angular Velocity and Rotation	18
5.3	Euler Angles	18
5.4	Moment of Inertia Tensor	19

5.5	Principal Axes	19
5.6	Parallel Axis Theorem	19
5.7	Angular Momentum	20
5.8	Kinetic Energy	20
5.9	Euler's Equations	21
5.10	Torque-Free Motion	21
5.11	Heavy Symmetric Top	21
5.12	Precession and Nutation	22
5.13	Practice Problems - Rigid Body Dynamics	22
6	Small Oscillations and Normal Modes	23
6.1	Motivation	23
6.2	Single Oscillator Review	23
6.3	Coupled Oscillators - Two Masses	23
6.4	General Theory of Small Oscillations	23
6.5	Normal Modes	24
6.6	Normal Coordinates	25
6.7	Orthogonality of Normal Modes	25
6.8	Energy in Normal Modes	26
6.9	Continuous Systems - String Vibrations	26
6.10	Damped Oscillations	26
6.11	Forced Oscillations and Resonance	27
6.12	Practice Problems - Small Oscillations	27
7	Classical Mechanics - GATE Exam Strategy	28
7.1	Topic-wise Weightage and Strategy	28
7.2	Common Problem Types	28
7.3	Essential Formula Quick Reference	29
7.4	Time-Saving Tips	29
7.5	Common Mistakes to Avoid	30
7.6	Must-Practice Problems	30
8	Final Words	31

1 Introduction

Important for GATE

Classical Mechanics is the most problem-solving intensive subject in GATE Physics. It requires not just understanding but the ability to apply concepts quickly and accurately. This guide covers every essential topic with complete problem-solving strategies and NO STONE LEFT UNTURNED.

1.1 Why Classical Mechanics is Crucial

1. Foundation for Quantum Mechanics and Statistical Mechanics
2. Most predictable question patterns in GATE
3. High weightage (15-20% of physics questions)
4. Directly applicable problem-solving skills
5. Tests mathematical physics mastery

1.2 Study Strategy

Exam Tip

- Master the formulation before solving problems
- Practice at least 50 problems per topic
- Time yourself - aim for 3-4 minutes per problem
- Understand physical meaning, not just equations
- Connect Lagrangian and Hamiltonian viewpoints

2 Lagrangian Formulation

2.1 Motivation and Fundamental Concepts

Definition 2.1. The **Lagrangian** of a system is defined as:

$$L = T - V$$

where T is kinetic energy and V is potential energy.

Important for GATE

Why Lagrangian Mechanics?

- Works with any set of coordinates (not just Cartesian)
- Constraints handled automatically
- Scalar function (easier than vector equations)
- Suitable for complex systems
- Basis for quantum field theory

2.2 Generalized Coordinates

Definition 2.2. Generalized coordinates $q_1, q_2, \dots, q_n$ are any set of independent coordinates that completely specify the configuration of a system.

Degrees of freedom: Number of independent generalized coordinates needed.

For N particles in 3D with k holonomic constraints:

$$\text{DOF} = 3N - k$$

Example 2.3. Find degrees of freedom for:

1. Single particle in space: $\text{DOF} = 3$ (can use x, y, z or r, θ, ϕ)
2. Particle on sphere: $\text{DOF} = 2$ (constraint: $x^2 + y^2 + z^2 = R^2$)
3. Double pendulum: $\text{DOF} = 2$ (use θ_1, θ_2)
4. Rigid body: $\text{DOF} = 6$ (3 translation + 3 rotation)

2.3 D'Alembert's Principle and Virtual Work

Definition 2.4. Virtual displacement δq_i : Infinitesimal displacement consistent with constraints at fixed time.

Virtual work:

$$\delta W = \sum_i F_i \cdot \delta r_i$$

Theorem 2.5 (D'Alembert's Principle). For a system in equilibrium under applied forces F_i and constraint forces:

$$\sum_i (F_i - \dot{p}_i) \cdot \delta r_i = 0$$

where constraint forces do no virtual work.

2.4 Euler-Lagrange Equations

Theorem 2.6 (Euler-Lagrange Equations). For a system with Lagrangian $L(q_i, \dot{q}_i, t)$, the equations of motion are:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

for each generalized coordinate q_i .

Important for GATE

Steps to Solve Problems Using Lagrangian:

1. Choose appropriate generalized coordinates
2. Write kinetic energy T in terms of q_i and $\dot{q}_i$
3. Write potential energy V in terms of q_i
4. Form Lagrangian: $L = T - V$
5. Apply Euler-Lagrange equation for each coordinate
6. Solve the resulting differential equations

Example 2.7. Simple Pendulum using Lagrangian Formulation

Setup: Mass m , length l , angle θ from vertical.

Solution:

1. Generalized coordinate: $q = \theta$

2. Kinetic energy:

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m(l\dot{\theta})^2 = \frac{1}{2}ml^2\dot{\theta}^2$$

3. Potential energy (taking lowest point as reference):

$$V = mgl(1 - \cos \theta)$$

4. Lagrangian:

$$L = T - V = \frac{1}{2}ml^2\dot{\theta}^2 - mgl(1 - \cos \theta)$$

5. Apply Euler-Lagrange:

$$\frac{\partial L}{\partial \theta} = -mgl \sin \theta, \quad \frac{\partial L}{\partial \dot{\theta}} = ml^2\dot{\theta}$$

$$\frac{d}{dt}(ml^2\dot{\theta}) = -mgl \sin \theta$$

6. Equation of motion:

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0$$

For small angles: $\ddot{\theta} + \omega_0^2 \theta = 0$ where $\omega_0 = \sqrt{g/l}$

2.5 Constraints

Definition 2.8. Holonomic constraints: Can be written as $f(q_1, \dots, q_n, t) = 0$

Examples:

- Particle on sphere: $x^2 + y^2 + z^2 - R^2 = 0$
- Rigid body: $|r_i - r_j| = \text{const}$
- Rolling without slipping: $x = R\theta$

Non-holonomic constraints: Involve velocities, cannot be integrated.

Example: Rolling without slipping in general: $v_{\text{contact}} = 0$

Exam Tip

Handling Constraints:

- **Holonomic:** Reduce number of coordinates OR use Lagrange multipliers
- **Non-holonomic:** Must use Lagrange multipliers

2.6 Lagrange Multipliers Method

Theorem 2.9. For constraints $g_k(q, t) = 0$, the modified Euler-Lagrange equations are:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = \sum_k \lambda_k \frac{\partial g_k}{\partial q_i}$$

where λ_k are Lagrange multipliers (related to constraint forces).

Example 2.10. Bead on Rotating Hoop

A bead of mass m slides on a hoop of radius R rotating with constant angular velocity ω about vertical diameter.

Solution: Generalized coordinate: θ (angle from top)

Position: $x = R \sin \theta \cos(\omega t)$, $y = R \sin \theta \sin(\omega t)$, $z = R \cos \theta$

Velocities:

$$\begin{aligned}\dot{x} &= R\dot{\theta} \cos \theta \cos(\omega t) - R\omega \sin \theta \sin(\omega t) \\ \dot{y} &= R\dot{\theta} \cos \theta \sin(\omega t) + R\omega \sin \theta \cos(\omega t) \\ \dot{z} &= -R\dot{\theta} \sin \theta\end{aligned}$$

Kinetic energy:

$$\begin{aligned}T &= \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) \\ &= \frac{1}{2}m[R^2\dot{\theta}^2 + R^2\omega^2 \sin^2 \theta]\end{aligned}$$

Potential energy: $V = -mgR \cos \theta$

Lagrangian:

$$L = \frac{1}{2}mR^2[\dot{\theta}^2 + \omega^2 \sin^2 \theta] + mgR \cos \theta$$

Euler-Lagrange equation:

$$\frac{d}{dt}(mR^2\dot{\theta}) - mR^2\omega^2 \sin\theta \cos\theta - mgR \sin\theta = 0$$

$$\ddot{\theta} = (\omega^2 \cos\theta - \frac{g}{R}) \sin\theta$$

Equilibrium: $\ddot{\theta} = 0 \implies \cos\theta = \frac{g}{R\omega^2}$ (for $\omega^2 > g/R$)

2.7 Generalized Momentum and Cyclic Coordinates

Definition 2.11. Generalized momentum (canonical momentum):

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

Cyclic (ignorable) coordinate: If $\frac{\partial L}{\partial q_i} = 0$, then q_i is cyclic.

Theorem 2.12. If q_i is a cyclic coordinate, then its conjugate momentum p_i is conserved:

$$\frac{dp_i}{dt} = 0 \implies p_i = \text{constant}$$

Important for GATE

Common Cyclic Coordinates and Conservation Laws:

- Translation invariance (x cyclic) $\implies$ Linear momentum conserved
- Rotational invariance (θ cyclic) $\implies$ Angular momentum conserved
- Time independence (t not explicit) $\implies$ Energy conserved

Example 2.13. Central Force Problem in Lagrangian Formulation

For a particle in central force $F(r)$ (potential $V(r)$):

Using polar coordinates (r, θ) :

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r)$$

θ is cyclic! Therefore:

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} = l = \text{const (angular momentum)}$$

For r :

$$\frac{d}{dt}(m\dot{r}) - mr\dot{\theta}^2 + \frac{dV}{dr} = 0$$

Using $l = mr^2\dot{\theta}$:

$$m\ddot{r} = \frac{l^2}{mr^3} - \frac{dV}{dr}$$

2.8 Conservation Laws and Noether's Theorem

Theorem 2.14 (Noether's Theorem). Every continuous symmetry of the Lagrangian corresponds to a conservation law:

- **Translation symmetry** $\implies$ Momentum conservation
- **Rotation symmetry** $\implies$ Angular momentum conservation
- **Time translation** $\implies$ Energy conservation

2.9 Energy Function and Conservation

Definition 2.15. The **energy function** (Jacobi's integral):

$$h = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L = \sum_i p_i \dot{q}_i - L$$

Theorem 2.16. If L does not depend explicitly on time ($\partial L / \partial t = 0$), then energy is conserved:

$$\frac{dh}{dt} = -\frac{\partial L}{\partial t}$$

If $\partial L / \partial t = 0$, then $h = E = T + V$ (total energy).

2.10 Practice Problems - Lagrangian Mechanics

1. Derive equations of motion for Atwood machine using Lagrangian.
2. Find Lagrangian for double pendulum and derive equations of motion.
3. A cylinder rolls without slipping down an incline. Use Lagrangian with constraint.
4. Particle on surface of sphere under gravity - find equation of motion.
5. Two masses connected by spring on frictionless table - find normal modes.

3 Hamiltonian Mechanics

3.1 From Lagrangian to Hamiltonian

Definition 3.1. The **Hamiltonian** is obtained via Legendre transformation:

$$H(q_i, p_i, t) = \sum_i p_i \dot{q}_i - L(q_i, \dot{q}_i, t)$$

where $p_i = \frac{\partial L}{\partial \dot{q}_i}$ are generalized momenta.

Important for GATE

Procedure to Find Hamiltonian:

1. Start with Lagrangian $L(q, \dot{q}, t)$
2. Find generalized momenta: $p_i = \frac{\partial L}{\partial \dot{q}_i}$
3. Invert to get $\dot{q}_i = \dot{q}_i(q, p, t)$
4. Substitute into $H = \sum p_i \dot{q}_i - L$
5. Express everything in terms of (q, p, t)

Example 3.2. Hamiltonian for Simple Harmonic Oscillator

Lagrangian: $L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$

Momentum: $p = \frac{\partial L}{\partial \dot{x}} = m\dot{x} \implies \dot{x} = \frac{p}{m}$

Hamiltonian:

$$\begin{aligned} H &= p\dot{x} - L = p \cdot \frac{p}{m} - \frac{1}{2}m \left(\frac{p}{m}\right)^2 + \frac{1}{2}kx^2 \\ &= \frac{p^2}{m} - \frac{p^2}{2m} + \frac{1}{2}kx^2 \\ &= \frac{p^2}{2m} + \frac{1}{2}kx^2 = T + V \end{aligned}$$

3.2 Hamilton's Equations

Theorem 3.3 (Hamilton's Canonical Equations).

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

These are $2n$ first-order equations (versus n second-order Euler-Lagrange equations).

Exam Tip

When to Use Hamiltonian vs Lagrangian:

- **Lagrangian:** Better for deriving equations of motion
- **Hamiltonian:** Better for phase space analysis, perturbation theory, quantum mechanics transition
- Both give same physics!

Example 3.4. Hamilton's Equations for SHO

From previous: $H = \frac{p^2}{2m} + \frac{1}{2}kx^2$

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m}$$

$$\dot{p} = -\frac{\partial H}{\partial x} = -kx$$

Combining: $m\ddot{x} = \dot{p} = -kx \implies \ddot{x} + \omega^2 x = 0$ where $\omega = \sqrt{k/m}$

3.3 Phase Space

Definition 3.5. Phase space is the $2n$ -dimensional space of all (q_i, p_i) coordinates.

Phase space trajectory: Path traced by system in phase space as it evolves.

Phase portrait: Collection of trajectories for different initial conditions.

Important for GATE**Properties of Phase Space Trajectories:**

1. Trajectories never cross (deterministic evolution)
2. Closed trajectory $\implies$ Periodic motion
3. Energy surfaces are constant H surfaces
4. Liouville's theorem: Phase space volume conserved

3.4 Poisson Brackets

Definition 3.6. The **Poisson bracket** of two functions $f(q, p, t)$ and $g(q, p, t)$ is:

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right)$$

Important for GATE**Fundamental Poisson Brackets (FPB):**

$$\{q_i, q_j\} = 0, \quad \{p_i, p_j\} = 0, \quad \{q_i, p_j\} = \delta_{ij}$$

Theorem 3.7 (Time Evolution via Poisson Brackets). For any function $f(q, p, t)$:

$$\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t}$$

In particular, Hamilton's equations are:

$$\dot{q}_i = \{q_i, H\}, \quad \dot{p}_i = \{p_i, H\}$$

Theorem 3.8 (Conservation and Poisson Brackets). A quantity f is conserved if:

$$\frac{df}{dt} = 0 \iff \{f, H\} = 0 \text{ and } \frac{\partial f}{\partial t} = 0$$

Example 3.9. Angular Momentum Conservation

For central force, $H = \frac{p^2}{2m} + V(r)$ where $r = \sqrt{x^2 + y^2 + z^2}$

Angular momentum: $L_z = xp_y - yp_x$

Check conservation:

$$\begin{aligned}\{L_z, H\} &= \{xp_y - yp_x, \frac{p_x^2 + p_y^2 + p_z^2}{2m} + V(r)\} \\ &= \{xp_y, V(r)\} - \{yp_x, V(r)\} \\ &= x \frac{\partial V}{\partial y} - y \frac{\partial V}{\partial x} \\ &= x \cdot \frac{dV}{dr} \cdot \frac{y}{r} - y \cdot \frac{dV}{dr} \cdot \frac{x}{r} = 0\end{aligned}$$

Therefore L_z is conserved!

3.5 Properties of Poisson Brackets

Formula 3.10 (Poisson Bracket Properties). 1. **Antisymmetry:** $\{f, g\} = -\{g, f\}$

2. **Linearity:** $\{af + bg, h\} = a\{f, h\} + b\{g, h\}$

3. **Leibniz rule:** $\{fg, h\} = f\{g, h\} + \{f, h\}g$

4. **Jacobi identity:** $\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$

3.6 Canonical Transformations

Definition 3.11. A transformation $(q, p) \rightarrow (Q, P)$ is **canonical** if it preserves the form of Hamilton's equations:

$$\dot{Q}_i = \frac{\partial K}{\partial P_i}, \quad \dot{P}_i = -\frac{\partial K}{\partial Q_i}$$

where $K(Q, P, t)$ is the new Hamiltonian.

Theorem 3.12 (Conditions for Canonical Transformation). A transformation is canonical if and only if:

1. It preserves Poisson brackets (symplectic condition):

$$\{Q_i, Q_j\} = 0, \quad \{P_i, P_j\} = 0, \quad \{Q_i, P_j\} = \delta_{ij}$$

2. OR there exists a generating function F such that:

$$p_i dq_i - P_i dQ_i = dF$$

3.7 Generating Functions

Important for GATE

Four Types of Generating Functions:

Type 1: $F_1(q, Q, t)$

$$p_i = \frac{\partial F_1}{\partial q_i}, \quad P_i = -\frac{\partial F_1}{\partial Q_i}, \quad K = H + \frac{\partial F_1}{\partial t}$$

Type 2: $F_2(q, P, t)$

$$p_i = \frac{\partial F_2}{\partial q_i}, \quad Q_i = \frac{\partial F_2}{\partial P_i}, \quad K = H + \frac{\partial F_2}{\partial t}$$

Type 3: $F_3(p, Q, t)$

$$q_i = -\frac{\partial F_3}{\partial p_i}, \quad P_i = -\frac{\partial F_3}{\partial Q_i}, \quad K = H + \frac{\partial F_3}{\partial t}$$

Type 4: $F_4(p, P, t)$

$$q_i = -\frac{\partial F_4}{\partial p_i}, \quad Q_i = \frac{\partial F_4}{\partial P_i}, \quad K = H + \frac{\partial F_4}{\partial t}$$

Example 3.13. Identity Transformation

$$F_2 = q_i P_i$$

$$p_i = \frac{\partial F_2}{\partial q_i} = P_i, \quad Q_i = \frac{\partial F_2}{\partial P_i} = q_i$$

This swaps coordinates and momenta!

Example 3.14. Point Transformation

$$F_2 = f(q, t)P \text{ generates } Q = f(q, t), \quad P = p \frac{\partial q}{\partial Q}$$

3.8 Hamilton-Jacobi Theory

Definition 3.15. The **Hamilton-Jacobi equation** is:

$$H\left(q_i, \frac{\partial S}{\partial q_i}, t\right) + \frac{\partial S}{\partial t} = 0$$

where $S(q, t)$ is Hamilton's principal function.

Theorem 3.16. If $S(q, \alpha, t)$ is a complete solution with n independent constants α_i , then:

$$p_i = \frac{\partial S}{\partial q_i}, \quad \beta_i = \frac{\partial S}{\partial \alpha_i}$$

give the solution with α_i and β_i as constants of integration.

Exam Tip**Hamilton-Jacobi Method:**

1. Write Hamilton-Jacobi equation
2. Try separation: $S = W(q) - Et$ for time-independent H
3. Solve for $W(q)$: $H(q, \partial W/\partial q) = E$
4. Find $p = \partial S/\partial q$ and integrate for $q(t)$

Example 3.17. Harmonic Oscillator via Hamilton-Jacobi

$$H = \frac{p^2}{2m} + \frac{1}{2}kx^2 = E$$

Try $S = W(x) - Et$:

$$\frac{1}{2m} \left(\frac{dW}{dx} \right)^2 + \frac{1}{2}kx^2 = E$$

$$\frac{dW}{dx} = \sqrt{2m(E - \frac{1}{2}kx^2)}$$

$$W = \int \sqrt{2mE - mkx^2} dx$$

Complete solution leads to $x(t) = A \cos(\omega t + \phi)$ where $\omega = \sqrt{k/m}$

3.9 Action-Angle Variables

Definition 3.18. For periodic motion, **action variable**:

$$J = \oint p dq = \oint p(q, E) dq$$

where integral is over one complete period.

Angle variable w conjugate to J increases by 2π per cycle.

Theorem 3.19. In action-angle variables:

$$H = H(J) \implies \dot{J} = 0, \quad \dot{w} = \frac{\partial H}{\partial J} = \omega(J)$$

Motion is uniform in w : $w = \omega t + w_0$

3.10 Practice Problems - Hamiltonian Mechanics

1. Find Hamiltonian for particle in electromagnetic field.
2. Calculate $\{L_x, L_y\}$ where $\vec{L} = \vec{r} \times \vec{p}$.
3. Show that $F_2 = qP \cot \theta$ generates canonical transformation.
4. Solve simple pendulum using Hamilton-Jacobi equation.
5. Find action-angle variables for 1D harmonic oscillator.

4 Central Force Motion

4.1 Central Force Definition

Definition 4.1. A central force is one that:

- Depends only on distance from origin: $\vec{F} = F(r)\hat{r}$
- Is along the radial direction
- Derives from potential: $F(r) = -\frac{dV}{dr}$

Important for GATE

Consequences of Central Force:

1. Angular momentum $\vec{L}$ is conserved
2. Motion is in a plane perpendicular to $\vec{L}$
3. Areal velocity is constant (Kepler's 2nd law)
4. Energy is conserved

4.2 Conservation Laws

Theorem 4.2 (Angular Momentum Conservation). For central force:

$$\vec{L} = \vec{r} \times \vec{p} = \text{constant}$$

Magnitude: $L = mr^2\dot{\theta}$

Theorem 4.3 (Areal Velocity). Area swept out per unit time:

$$\frac{dA}{dt} = \frac{1}{2}r^2\dot{\theta} = \frac{L}{2m} = \text{constant}$$

4.3 Equation of Motion in Plane

Formula 4.4 (Central Force Equations). In polar coordinates (r, θ) :

Radial equation:

$$m\ddot{r} = mr\dot{\theta}^2 - \frac{dV}{dr} = \frac{L^2}{mr^3} - \frac{dV}{dr}$$

Angular equation:

$$\frac{d}{dt}(mr^2\dot{\theta}) = 0 \implies L = mr^2\dot{\theta} = \text{const}$$

4.4 Effective Potential

Definition 4.5. The **effective potential** is:

$$V_{\text{eff}}(r) = V(r) + \frac{L^2}{2mr^2}$$

where the second term is the centrifugal barrier.

Important for GATE

Energy Equation:

$$E = \frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r)$$

This reduces 2D motion to effective 1D motion in r !

Analysis:

- $E > V_{\text{eff}}$: Motion possible
- $E = V_{\text{eff}}$: Turning point
- $\frac{dV_{\text{eff}}}{dr} = 0$ and $E = V_{\text{eff}}$: Circular orbit

4.5 Orbit Equation

Theorem 4.6 (Binet's Equation). Let $u = 1/r$. The orbit equation is:

$$\frac{d^2u}{d\theta^2} + u = -\frac{m}{L^2u^2}F(1/u)$$

For potential $V(r)$:

$$\frac{d^2u}{d\theta^2} + u = -\frac{m}{L^2u^2} \frac{d}{dr}V(r) = \frac{m}{L^2} \frac{d}{du}V(1/u)$$

4.6 Kepler Problem (Inverse Square Law)

Definition 4.7. Gravitational force: $F = -\frac{GMm}{r^2} = -\frac{k}{r^2}$

Potential: $V(r) = -\frac{k}{r}$ where $k = GMm$

4.6.1 Orbit Equation for Kepler Problem

Theorem 4.8. For inverse square law, Binet's equation gives:

$$\frac{d^2u}{d\theta^2} + u = \frac{mk}{L^2}$$

General solution:

$$u = \frac{1}{r} = \frac{mk}{L^2}(1 + e \cos(\theta - \theta_0))$$

Or, choosing $\theta_0 = 0$:

$$r = \frac{l}{1 + e \cos \theta}$$

where $l = \frac{L^2}{mk}$ is the semi-latus rectum and e is eccentricity.

Important for GATE**Conic Sections Based on Eccentricity:**

- $e = 0$: Circle (constant r)
- $0 < e < 1$: Ellipse (bound orbit)
- $e = 1$: Parabola (just unbound)
- $e > 1$: Hyperbola (unbound, scattering)

Eccentricity in terms of energy:

$$e = \sqrt{1 + \frac{2EL^2}{mk^2}}$$

4.6.2 Kepler's Laws

Theorem 4.9 (Kepler's Three Laws). **1. Law of Orbits:** Planets move in ellipses with Sun at one focus.

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta}$$

2. Law of Areas: Equal areas swept in equal times.

$$\frac{dA}{dt} = \frac{L}{2m} = \text{const}$$

3. Law of Periods: $T^2 \propto a^3$

$$T^2 = \frac{4\pi^2}{GM} a^3$$

where a is semi-major axis.

Example 4.10. Derive Kepler's Third Law

For ellipse: $a = \frac{l}{1-e^2}$, Area = $\pi ab = \pi a^2 \sqrt{1-e^2}$

Areal velocity: $\frac{dA}{dt} = \frac{L}{2m}$

Period: $T = \frac{\text{Area}}{dA/dt} = \frac{\pi a^2 \sqrt{1-e^2}}{L/2m}$

Using $l = L^2/(mk)$ and $a = l/(1-e^2)$:

$$T^2 = \frac{4\pi^2 m a^3}{k} = \frac{4\pi^2 a^3}{GM}$$

4.7 Energy Analysis

Formula 4.11 (Energy for Different Orbits). Total energy: $E = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - \frac{k}{r}$

Using $L = mr^2\dot{\theta}$:

$$E = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} - \frac{k}{r}$$

For circular orbit ($\dot{r} = 0$, $r = r_0$):

$$E = \frac{L^2}{2mr_0^2} - \frac{k}{r_0} = -\frac{k}{2r_0} \quad (\text{using force balance})$$

For elliptical orbit:

$$E = -\frac{k}{2a}$$

where a is semi-major axis.

For parabolic orbit: $E = 0$

For hyperbolic orbit: $E > 0$

4.8 Scattering in Central Force

Definition 4.12. For repulsive central force (e.g., Coulomb scattering), particles follow hyperbolic trajectories.

Impact parameter b : Perpendicular distance from force center to incoming trajectory.

Scattering angle Θ : Angle between incoming and outgoing asymptotes.

Formula 4.13 (Rutherford Scattering). For Coulomb potential $V = k/r$ (repulsive, $k > 0$):

$$\Theta = \pi - 2\theta_{\min}$$

where $\theta_{\min}$ is angle at closest approach.

Relation between impact parameter and scattering angle:

$$b = \frac{k}{2E} \cot \frac{\Theta}{2}$$

Differential cross-section:

$$\frac{d\sigma}{d\Omega} = \left(\frac{k}{4E} \right)^2 \frac{1}{\sin^4(\Theta/2)}$$

4.9 Virial Theorem

Theorem 4.14 (Virial Theorem). For a bound system with potential $V = kr^n$:

$$\langle T \rangle = \frac{n}{2} \langle V \rangle$$

For inverse square law ($n = -1$):

$$\langle T \rangle = -\frac{1}{2} \langle V \rangle \implies E = \langle T \rangle + \langle V \rangle = \frac{1}{2} \langle V \rangle = -\langle T \rangle$$

4.10 Practice Problems - Central Force Motion

1. Particle in potential $V = -k/r + c/r^2$. Find conditions for circular orbit.
2. Calculate escape velocity from Earth's surface.
3. For elliptical orbit, find ratio of speeds at perihelion and aphelion.
4. Derive scattering angle for hard sphere scattering.
5. Find period of small radial oscillations about circular orbit.

5 Rigid Body Dynamics

5.1 Rigid Body Definition and Kinematics

Definition 5.1. A rigid body is a collection of particles with fixed distances between them:

$$|r_i - r_j| = \text{constant}$$

Degrees of freedom: 6 (3 translation + 3 rotation)

5.2 Angular Velocity and Rotation

Definition 5.2. Angular velocity vector $\vec{\omega}$:

Velocity of point P relative to origin:

$$\vec{v} = \vec{v}_{\text{CM}} + \vec{\omega} \times \vec{r}'$$

where $\vec{r}'$ is position relative to center of mass.

Formula 5.3 (Rotation Matrices). Rotation about z -axis by angle ϕ :

$$R_z(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Similarly for $R_x(\theta)$ and $R_y(\psi)$.

5.3 Euler Angles

Definition 5.4. Euler angles (ϕ, θ, ψ) specify orientation:

1. Rotation by ϕ about z -axis
2. Rotation by θ about new x -axis (line of nodes)
3. Rotation by ψ about new z -axis

Total rotation: $R = R_z(\phi)R_x(\theta)R_z(\psi)$

Important for GATE

Angular Velocity in Terms of Euler Angles:

$$\omega_x = \dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi \quad (1)$$

$$\omega_y = \dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi \quad (2)$$

$$\omega_z = \dot{\phi} \cos \theta + \dot{\psi} \quad (3)$$

5.4 Moment of Inertia Tensor

Definition 5.5. The **moment of inertia tensor** is:

$$I_{ij} = \sum_{\alpha} m_{\alpha} (r_{\alpha}^2 \delta_{ij} - x_{i\alpha} x_{j\alpha})$$

In matrix form:

$$\mathbf{I} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix}$$

where:

$$I_{xx} = \sum m(y^2 + z^2) \quad (4)$$

$$I_{yy} = \sum m(x^2 + z^2) \quad (5)$$

$$I_{zz} = \sum m(x^2 + y^2) \quad (6)$$

$$I_{xy} = I_{yx} = - \sum mxy \quad (\text{products of inertia}) \quad (7)$$

Important for GATE

For Continuous Body:

$$I_{xx} = \int (y^2 + z^2) \rho dV, \quad I_{xy} = - \int xy \rho dV$$

5.5 Principal Axes

Theorem 5.6. Every rigid body has three orthogonal **principal axes** along which $\mathbf{I}$ is diagonal:

$$\mathbf{I} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

Principal moments are eigenvalues of inertia tensor.

Exam Tip

Finding Principal Axes:

1. Form inertia tensor $\mathbf{I}$
2. Solve $\det(\mathbf{I} - I\lambda) = 0$ for eigenvalues I_1, I_2, I_3
3. Find eigenvectors (principal axes)

5.6 Parallel Axis Theorem

Theorem 5.7 (Parallel Axis Theorem). If I_{CM} is moment of inertia about axis through center of mass, and I is moment about parallel axis at distance d :

$$I = I_{\text{CM}} + Md^2$$

where M is total mass.

Theorem 5.8 (Perpendicular Axis Theorem). For planar body in xy -plane:

$$I_z = I_x + I_y$$

Example 5.9. Moments of Inertia for Common Shapes

1. Uniform Rod (length L , mass M):

- About center, perpendicular: $I = \frac{1}{12}ML^2$
- About end, perpendicular: $I = \frac{1}{3}ML^2$

2. Uniform Disk (radius R , mass M):

- About center, perpendicular to disk: $I = \frac{1}{2}MR^2$
- About diameter: $I = \frac{1}{4}MR^2$

3. Solid Sphere (radius R , mass M):

- About diameter: $I = \frac{2}{5}MR^2$

4. Hollow Sphere (radius R , mass M):

- About diameter: $I = \frac{2}{3}MR^2$

5. Solid Cylinder (radius R , length L , mass M):

- About axis: $I = \frac{1}{2}MR^2$
- About diameter through center: $I = \frac{1}{12}ML^2 + \frac{1}{4}MR^2$

5.7 Angular Momentum

Definition 5.10. For rigid body rotating with $\vec{\omega}$:

$$\vec{L} = \mathbf{I} \cdot \vec{\omega}$$

In component form: $L_i = \sum_j I_{ij}\omega_j$

Important for GATE

Key Point: $\vec{L}$ and $\vec{\omega}$ are NOT necessarily parallel unless rotating about principal axis!

In principal axes:

$$L_1 = I_1\omega_1, \quad L_2 = I_2\omega_2, \quad L_3 = I_3\omega_3$$

5.8 Kinetic Energy

Formula 5.11 (Kinetic Energy of Rigid Body).

$$T = \frac{1}{2}Mv_{\text{CM}}^2 + \frac{1}{2}\vec{\omega} \cdot \mathbf{I} \cdot \vec{\omega}$$

In principal axes:

$$T_{\text{rot}} = \frac{1}{2}(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2)$$

5.9 Euler's Equations

Theorem 5.12 (Euler's Equations of Motion). In body-fixed principal axes:

$$I_1 \dot{\omega}_1 + (I_3 - I_2) \omega_2 \omega_3 = N_1 \quad (8)$$

$$I_2 \dot{\omega}_2 + (I_1 - I_3) \omega_3 \omega_1 = N_2 \quad (9)$$

$$I_3 \dot{\omega}_3 + (I_2 - I_1) \omega_1 \omega_2 = N_3 \quad (10)$$

where N_i are components of external torque.

Exam Tip

These equations account for the fact that principal axes rotate with the body!

5.10 Torque-Free Motion

Theorem 5.13 (Torque-Free Symmetric Top). For symmetric top ($I_1 = I_2 \neq I_3$) with no external torque:

$$\omega_3 = \text{constant}$$

ω_1 and ω_2 precess:

$$\omega_1 = A \cos(\Omega t), \quad \omega_2 = A \sin(\Omega t)$$

where $\Omega = \frac{I_3 - I_1}{I_1} \omega_3$

5.11 Heavy Symmetric Top

Definition 5.14. Symmetric top with one point fixed (e.g., spinning top under gravity).

Using Euler angles, Lagrangian is:

$$L = \frac{1}{2} I_1 (\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2} I_3 (\dot{\psi} + \dot{\phi} \cos \theta)^2 - Mgl \cos \theta$$

Important for GATE

Constants of Motion:

- Energy E (time-independent)
- $p_\phi = L_z$ (azimuthal angular momentum, ϕ cyclic)
- $p_\psi = I_3 \omega_3$ (spin angular momentum, ψ cyclic)

These lead to:

$$\frac{1}{2} I_1 \dot{\theta}^2 + V_{\text{eff}}(\theta) = E$$

where

$$V_{\text{eff}}(\theta) = \frac{(p_\phi - p_\psi \cos \theta)^2}{2I_1 \sin^2 \theta} + Mgl \cos \theta$$

5.12 Precession and Nutation

Definition 5.15. Precession: Slow rotation of symmetry axis around vertical ($\dot{\phi}$)

Nutation: Wobbling of symmetry axis (θ oscillation)

Spin: Fast rotation about symmetry axis ($\dot{\psi}$)

Formula 5.16 (Steady Precession). For fast-spinning top (ω_3 large), approximate steady precession rate:

$$\Omega_p = \dot{\phi} \approx \frac{Mgl}{I_3\omega_3}$$

5.13 Practice Problems - Rigid Body Dynamics

1. Find moment of inertia tensor for rectangular plate.
2. Solid sphere rolls without slipping down incline - find acceleration.
3. Symmetric top spinning - derive Euler's equations and find solutions.
4. Calculate precession frequency of gyroscope.
5. Torque-free asymmetric top - discuss stability of rotation about each principal axis.

6 Small Oscillations and Normal Modes

6.1 Motivation

Important for GATE

Small oscillations about equilibrium appear everywhere:

- Molecular vibrations
- Crystal lattice vibrations (phonons)
- Structural engineering
- Quantum harmonic oscillators
- Perturbations in orbits

Method: Expand potential near equilibrium and keep quadratic terms.

6.2 Single Oscillator Review

Formula 6.1 (Simple Harmonic Oscillator). Equation: $m\ddot{x} + kx = 0$ or $\ddot{x} + \omega_0^2 x = 0$ where $\omega_0 = \sqrt{k/m}$

Solution: $x(t) = A \cos(\omega_0 t + \phi)$

Energy: $E = \frac{1}{2}kA^2 = \frac{1}{2}m\omega_0^2 A^2$

6.3 Coupled Oscillators - Two Masses

Example 6.2. Two Masses Connected by Springs

Setup: Masses m_1, m_2 connected by springs k_1, k, k_2 on frictionless surface.

Displacements: x_1, x_2 from equilibrium.

Equations of motion:

$$m_1 \ddot{x}_1 = -k_1 x_1 + k(x_2 - x_1) \quad (11)$$

$$m_2 \ddot{x}_2 = -k(x_2 - x_1) - k_2 x_2 \quad (12)$$

Matrix form:

$$\begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix} \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} = - \begin{pmatrix} k_1 + k & -k \\ -k & k + k_2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Or: $M\ddot{\vec{x}} = -K\vec{x}$

6.4 General Theory of Small Oscillations

Theorem 6.3 (Small Oscillations About Equilibrium). For system with generalized coordinates q_i , expand potential about equilibrium $q_i^{(0)}$:

$$V = V_0 + \sum_i \left. \frac{\partial V}{\partial q_i} \right|_0 \eta_i + \frac{1}{2} \sum_{ij} \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_0 \eta_i \eta_j + \dots$$

where $\eta_i = q_i - q_i^{(0)}$ are displacements.

At equilibrium: $\left. \frac{\partial V}{\partial q_i} \right|_0 = 0$

Keeping quadratic terms:

$$V \approx V_0 + \frac{1}{2} \sum_{ij} V_{ij} \eta_i \eta_j$$

where $V_{ij} = \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_0$

Formula 6.4 (Kinetic Energy). For small velocities:

$$T = \frac{1}{2} \sum_{ij} T_{ij} \dot{\eta}_i \dot{\eta}_j$$

where T_{ij} are mass matrix elements (often $T_{ij} = m_i \delta_{ij}$).

6.5 Normal Modes

Definition 6.5. A **normal mode** is a motion where all parts oscillate with same frequency and phase.

Try solution: $\eta_i(t) = a_i e^{i\omega t}$

Substituting into $T\ddot{\vec{\eta}} + V\vec{\eta} = 0$:

$$(-\omega^2 \mathbf{T} + \mathbf{V})\vec{a} = 0$$

Non-trivial solution requires:

$$\det(\mathbf{V} - \omega^2 \mathbf{T}) = 0$$

This is the **secular equation** giving n normal frequencies $\omega_1, \dots, \omega_n$.

Important for GATE

Procedure to Find Normal Modes:

1. Set up kinetic and potential energy matrices $\mathbf{T}$ and $\mathbf{V}$
2. Solve secular equation $\det(\mathbf{V} - \omega^2 \mathbf{T}) = 0$ for eigenvalues ω_k^2
3. For each ω_k , solve $(\mathbf{V} - \omega_k^2 \mathbf{T})\vec{a}^{(k)} = 0$ for eigenvector $\vec{a}^{(k)}$
4. General solution is superposition:

$$\eta_i(t) = \sum_k C_k a_i^{(k)} \cos(\omega_k t + \phi_k)$$

5. Determine C_k and ϕ_k from initial conditions

Example 6.6. Two Identical Masses, Three Identical Springs

$$m_1 = m_2 = m, k_1 = k = k_2$$

$$\text{Mass matrix: } \mathbf{M} = m\mathbf{I}$$

$$\text{Stiffness matrix: } \mathbf{K} = k \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Secular equation:

$$\begin{aligned} \det \begin{pmatrix} 2k - m\omega^2 & -k \\ -k & 2k - m\omega^2 \end{pmatrix} &= 0 \\ (2k - m\omega^2)^2 - k^2 &= 0 \\ 2k - m\omega^2 &= \pm k \end{aligned}$$

Normal frequencies:

$$\omega_1 = \sqrt{\frac{k}{m}}, \quad \omega_2 = \sqrt{\frac{3k}{m}}$$

$$\text{For } \omega_1: \vec{a}^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ (symmetric mode - masses move together)}$$

$$\text{For } \omega_2: \vec{a}^{(2)} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ (antisymmetric mode - masses move opposite)}$$

General solution:

$$x_1(t) = A_1 \cos(\omega_1 t + \phi_1) + A_2 \cos(\omega_2 t + \phi_2) \quad (13)$$

$$x_2(t) = A_1 \cos(\omega_1 t + \phi_1) - A_2 \cos(\omega_2 t + \phi_2) \quad (14)$$

6.6 Normal Coordinates

Definition 6.7. Normal coordinates Q_k are linear combinations of η_i such that Lagrangian becomes:

$$L = \frac{1}{2} \sum_k (\dot{Q}_k^2 - \omega_k^2 Q_k^2)$$

Each Q_k oscillates independently:

$$\ddot{Q}_k + \omega_k^2 Q_k = 0$$

Formula 6.8 (Transformation to Normal Coordinates).

$$\eta_i = \sum_k a_i^{(k)} Q_k$$

where $\vec{a}^{(k)}$ are normalized eigenvectors.

In matrix form: $\vec{\eta} = \mathbf{A}\vec{Q}$ where columns of $\mathbf{A}$ are eigenvectors.

6.7 Orthogonality of Normal Modes

Theorem 6.9. Normal modes satisfy orthogonality relations:

$$\sum_i T_{ij} a_j^{(k)} a_i^{(l)} = M_k \delta_{kl}$$

$$\sum_i V_{ij} a_j^{(k)} a_i^{(l)} = M_k \omega_k^2 \delta_{kl}$$

For simple case $T_{ij} = m\delta_{ij}$:

$$\vec{a}^{(k)} \cdot \vec{a}^{(l)} = \delta_{kl} \quad (\text{after normalization})$$

6.8 Energy in Normal Modes

Formula 6.10. In normal coordinates:

$$E = \sum_k E_k = \sum_k \frac{1}{2}(\dot{Q}_k^2 + \omega_k^2 Q_k^2)$$

Each normal mode carries energy independently:

$$E_k = \frac{1}{2}\omega_k^2 A_k^2$$

where $Q_k(t) = A_k \cos(\omega_k t + \phi_k)$.

6.9 Continuous Systems - String Vibrations

Example 6.11. Vibrating String (Length L , Tension T , Linear Density μ)

Wave equation:

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$$

where $v = \sqrt{T/\mu}$ is wave speed.

Boundary conditions: $y(0, t) = y(L, t) = 0$

Normal modes:

$$y_n(x, t) = A_n \sin\left(\frac{n\pi x}{L}\right) \cos(\omega_n t + \phi_n)$$

Normal frequencies:

$$\omega_n = \frac{n\pi v}{L} = \frac{n\pi}{L} \sqrt{\frac{T}{\mu}}, \quad n = 1, 2, 3, \dots$$

General solution:

$$y(x, t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) \cos(\omega_n t + \phi_n)$$

6.10 Damped Oscillations

Formula 6.12 (Damped Harmonic Oscillator). Equation: $m\ddot{x} + \gamma\dot{x} + kx = 0$ or $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$

where $\beta = \gamma/(2m)$ is damping coefficient, $\omega_0 = \sqrt{k/m}$.

Three cases:

1. Underdamped ($\beta < \omega_0$): Oscillatory decay

$$x(t) = Ae^{-\beta t} \cos(\omega t + \phi), \quad \omega = \sqrt{\omega_0^2 - \beta^2}$$

2. Critically damped ($\beta = \omega_0$): Fastest return without oscillation

$$x(t) = (A + Bt)e^{-\beta t}$$

3. Overdamped ($\beta > \omega_0$): Slow exponential decay

$$x(t) = Ae^{-(\beta - \sqrt{\beta^2 - \omega_0^2})t} + Be^{-(\beta + \sqrt{\beta^2 - \omega_0^2})t}$$

6.11 Forced Oscillations and Resonance

Formula 6.13 (Driven Damped Oscillator). Equation: $m\ddot{x} + \gamma\dot{x} + kx = F_0 \cos(\omega_d t)$

Steady-state solution:

$$x(t) = A(\omega_d) \cos(\omega_d t - \delta)$$

Amplitude:

$$A(\omega_d) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + (2\beta\omega_d)^2}}$$

Phase lag:

$$\tan \delta = \frac{2\beta\omega_d}{\omega_0^2 - \omega_d^2}$$

Resonance: Maximum amplitude at $\omega_d = \omega_r = \sqrt{\omega_0^2 - 2\beta^2}$

At resonance:

$$A_{\max} = \frac{F_0}{2\beta m \omega_0} \approx \frac{F_0}{2\beta m \omega_r} \quad (\text{for small } \beta)$$

Quality factor:

$$Q = \frac{\omega_0}{2\beta} = \frac{\text{Energy stored}}{\text{Energy lost per cycle}}$$

6.12 Practice Problems - Small Oscillations

1. Three identical masses connected by identical springs in a line. Find all normal frequencies and modes.
2. Double pendulum with small angles - find normal modes.
3. Molecule vibrations: CO₂ (linear) - find normal modes (3 modes).
4. Loaded string: N equal masses on string - find normal frequencies (limit to continuous string).
5. Driven oscillator: find width of resonance peak (full width at half maximum).

7 Classical Mechanics - GATE Exam Strategy

7.1 Topic-wise Weightage and Strategy

Exam Tip

Expected Question Distribution:

- **Lagrangian/Hamiltonian:** 2-3 questions (High weightage!)
- **Central Force/Kepler:** 1-2 questions
- **Rigid Body:** 1-2 questions
- **Small Oscillations:** 1-2 questions

7.2 Common Problem Types

1. **Direct Lagrangian formulation:** Given system, write L and derive equations
2. **Constrained motion:** Pendulums, beads on wires, rolling
3. **Central force orbits:** Find eccentricity, period, energy
4. **Moment of inertia calculations:** Parallel axis, various shapes
5. **Normal mode frequencies:** Coupled oscillators, find eigenvalues

7.3 Essential Formula Quick Reference

Important for GATE

Lagrangian Mechanics:

$$L = T - V$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

Hamiltonian Mechanics:

$$H = \sum p_i \dot{q}_i - L$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right)$$

Central Force:

$$r = \frac{l}{1 + e \cos \theta} \quad (\text{orbit equation})$$

$$E = -\frac{k}{2a} \quad (\text{ellipse energy})$$

$$T^2 = \frac{4\pi^2}{GM} a^3 \quad (\text{Kepler's 3rd law})$$

Rigid Body:

$$\vec{L} = \mathbf{I} \cdot \vec{\omega}$$

$$T_{\text{rot}} = \frac{1}{2} \vec{\omega} \cdot \mathbf{I} \cdot \vec{\omega}$$

$$I = I_{\text{CM}} + Md^2 \quad (\text{parallel axis})$$

Small Oscillations:

$$\det(\mathbf{V} - \omega^2 \mathbf{M}) = 0 \quad (\text{secular equation})$$

$$\omega_{\text{res}} = \sqrt{\omega_0^2 - 2\beta^2} \quad (\text{resonance})$$

7.4 Time-Saving Tips

1. Always check for cyclic coordinates first - conserved momenta save time!
2. For central force, remember L is conserved - reduces to 1D problem
3. Know moments of inertia by heart for standard shapes

4. For coupled oscillators, look for symmetry - can simplify eigenvalue problem
5. In exams, you can often guess normal mode patterns from symmetry

7.5 Common Mistakes to Avoid

1. Forgetting minus sign in $L = T - V$ (not $T + V$!)
2. Wrong sign in Hamilton's equations ($\dot{p} = -\partial H/\partial q$)
3. Confusing angular momentum L with angular velocity ω
4. Not using parallel axis theorem when needed
5. Wrong normalization of eigenvectors
6. Forgetting $r^2\dot{\theta}$ term in polar coordinates kinetic energy

7.6 Must-Practice Problems

1. Bead on rotating hoop/wire
2. Double pendulum
3. Particle in central potential with $V = -k/r + c/r^2$
4. Atwood machine variations
5. Rolling cylinders/spheres
6. Symmetric top under gravity
7. Three-mass-three-spring system
8. Forced oscillator at resonance

8 Final Words

Classical Mechanics is the most problem-solving intensive subject. Success requires:

1. **Deep Understanding:** Know why each formulation works
2. **Pattern Recognition:** Most problems are variations of standard types
3. **Speed:** Practice until you can set up Lagrangian in under 1 minute
4. **Accuracy:** Double-check signs and factors of 2!

Important for GATE

30-Day Study Plan:

- **Week 1:** Lagrangian formulation - 20 problems
- **Week 2:** Hamiltonian mechanics - 20 problems
- **Week 3:** Central force + Rigid body - 15 + 15 problems
- **Week 4:** Small oscillations + Mixed practice - 20 problems

Last 10 Days: Previous year GATE problems only!

**Master Classical Mechanics = Strong Foundation for
All of Physics!**

All the Best for GATE 2026!